

Java Programming

Practice Programs

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1. Matrix Multiplication:

Q1 Matrix multiplication

Matrix multiplication

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int r1 = sc.nextInt();
7         int c1 = sc.nextInt();
8
9         int r2 = sc.nextInt();
10        int c2 = sc.nextInt();
11
12        if (c1 != r2) {
13            System.out.println("Matrix multiplication not possible");
14            return;
15        }
16
17        int[][] A = new int[r1][c1];
```

```
16
17        int[][] A = new int[r1][c1];
18        int[][] B = new int[r2][c2];
19        int[][] C = new int[r1][c2];
20
21
22        for (int i = 0; i < r1; i++) {
23            for (int j = 0; j < c1; j++) {
24                A[i][j] = sc.nextInt();
25            }
26        }
27
28        for (int i = 0; i < r2; i++) {
29            for (int j = 0; j < c2; j++) {
30                B[i][j] = sc.nextInt();
31            }
32        }
33        for (int i = 0; i < r1; i++) {
34            for (int j = 0; j < c2; j++) {
35                for (int k = 0; k < c1; k++) {
36                    C[i][j] += A[i][k] * B[k][j];
37                }
38            }
39        }
40        for (int i = 0; i < r1; i++) {
```

```

29         for (int j = 0; j < c2; j++) {
30             B[i][j] = sc.nextInt();
31         }
32     }
33     for (int i = 0; i < r1; i++) {
34         for (int j = 0; j < c2; j++) {
35             for (int k = 0; k < c1; k++) {
36                 C[i][j] += A[i][k] * B[k][j];
37             }
38         }
39     }
40     for (int i = 0; i < r1; i++) {
41         for (int j = 0; j < c2; j++) {
42             System.out.print(C[i][j] + " ");
43         }
44         System.out.println();
45     }
46
47     sc.close();
48 }
49 }

```

Input

```

2 2
2 2
1 2
3 4

```

Output

```

19 22
43 50

```

2. Largest Numbers :

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int max = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] > max)
17                max = a[i];
18        }
19
20        System.out.println("Largest element = " + max);
21    }
22 }

```

Input

```
5
10 25 8 40 15
```

Output

```
Largest element = 40
```

3. Smallest Numbers :

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int min = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] < min)
17                min = a[i];
18        }
19
20        System.out.println("Smallest element = " + min);
21    }
22 }
```

Input

```
5
10 20 6 35 50
```

Output

```
Smallest element = 6
```

4. NTH Largest Number :

```
1 import java.util.Scanner;
2 import java.util.Arrays;
3
4 public class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int n = sc.nextInt();
9         int[] a = new int[n];
10
11         for (int i = 0; i < n; i++)
12             a[i] = sc.nextInt();
13
14         int k = sc.nextInt();
15
16         Arrays.sort(a);
17
18         System.out.println(a[n - k]);
19     }
20 }
```

Input

5
10 25 8 40 15
2

Output

25

5. Plindrome number/String:

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
        int[] a = new int[20];  
        int count = 0, temp = n;  
  
        while (temp > 0) {  
            a[count++] = temp % 10;  
            temp /= 10;  
        }  
  
        boolean palindrome = true;  
        for (int i = 0; i < count / 2; i++) {  
            if (a[i] != a[count - 1 - i]) {  
                palindrome = false;  
                break;  
            }  
        }  
  
        if (palindrome)  
            System.out.println("Palindrome");  
        else  
            System.out.println("Not Palindrome");  
    }  
}
```

Input

121

Output

Palindrome

6. Removing Duplicates in array:

```
1 import java.util.Scanner;|
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5
6         int n = sc.nextInt();
7         int[] a = new int[n];
8
9         for (int i = 0; i < n; i++)
10             a[i] = sc.nextInt();
11
12         for (int i = 0; i < n; i++) {
13             boolean duplicate = false;
14
15             for (int j = 0; j < i; j++) {
16                 if (a[i] == a[j]) {
17                     duplicate = true;
18                     break;
19                 }
20             }
21
22             if (!duplicate)
23                 System.out.print(a[i] + " ");
24         }
```

Input

7
10 20 10 30 20 40 50

Output

10 20 30 40 50